



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -2

Deep Machine Learning in Time Series: Cardiac and Pulmonary Signals analysis using CNN models

DATE & TIME:
13th January 2021,
4:00 PM.

PLATFORM & WEB-LINK
Google Meet
meet.google.com/uhm-ykwu-ejn



SPEAKER
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ABSTRACT

A time series is a series of physical quantity varies over time. It has a wide range of applications. The proposed system involves a convolutional neural network (1D-CNN) model for time series data because of its high accuracy and robustness. Cardiovascular and pulmonary diseases are critical diseases and need to be diagnosed as early as possible. There is a lack of medical professionals in remote areas to diagnose these diseases. This work presents an automatic method using deep learning to diagnose the cardiac disorders from the PCG signal and lungs disorder from Pulmonary signals. To improve the accuracy in a noisy environment and make the method robust, the proposed method has used data augmentation techniques for training and multi-classification of multiple cardiac diseases. The proposed model can be ported small computing devices to make a stand-alone diagnostic tool that may be of help in remote primary health care centers. The proposed method is non-invasive, efficient, robust, and has low time complexity making it suitable for real-time applications.